Nandi Seed Recommender

February 20, 2026

1 Introduction

2 Datasets

2.1 Soil and Weather Data

Soil data are primarily based on iSDAsoil (Hengl *et al.*, 2021), which provides 21 structural and nutrient-based indicators (of which 17 are used) at 30m resolution for depths of 0-20cm and 20-50cm, alongside associated accuracy and per-pixel uncertainty.[1] For elevation and slope measurements, 30m-resolution NASA SRTM data is used.[2] Temperature and humidity data are obtained from ERA5 ($\sim$ 9km resolution) while CHIRPS is used for precipitation ($\sim$ 5km); both are available at daily resolution.[3][4]

We define the *Long Rains* season as March 25 to July 13 each year, based on an average 110-day growing season [5] and a reported optimal planting time in Nandi of March 25 to April 20.[6] Similarly, *Short Rains* is defined from October 25 to February 12, based on an optimal planting time of October 25 to November 15.[6] The ten-year mean (2015-2024) over these dates is calculated to obtain Mean Season Rainfall and Mean Temperature. The ten past years - rather than a longer average - is selected to more faithfully represent recent climate trends. The same process is applied to by-month precipitation (for each of the four months of the growing season), and for Mean Minimum and Mean Maximum Temperatures. To account for variation in planting times across the ideal window and germination time [7], Mean Germination Temperature is calculated via a 10-year average of the mean temperature over the 30 days following March 25 and October 25.

Droughts and water logging after periods of high rainfall are accounted for by employing the CropSuite threshold constraints in Table 1. With respect to drought, for each year of 1981-2024 and for a given 9km x 9km pixel, we iterate through all 7-day periods over the growing season. If any period exhibits a temperature of below 5°C for seven consecutive days, that pixel is classified as a crop failure for that year. The Crop Failure Frequency is then calculated by division of the number of failure years by total number of years. The same process is applied to high temperatures and high and low precipitation values, as per CropSuite Threshold Contraints in Table 1.[5] RONI data is extracted from `https://www.cpc.ncep.noaa.gov/data/indices/RONI.ascii.txt` for MAM (long rains) and OND (short rains) for 1950-2024 and classified as El Nino (EN), La Nina (LN), or Neutral (N).

- How to incorporate drainage? NB: We could also calculate over 5 months and then do these calculations real time so the user could choose their planting date.

2.2 Suitability Data

Maize land suitability parameters for climate, terrain, soil physical properties, and nutrient concentrations are predominantly derived from Sys *et al.* (1993), as is common practice in existing literature.[8][9][10][5] Here, suitability is divided into four main classes, with six total subclasses, as detailed in Table 1.[11] Each class is associated with an associated "Suitability Score", ranging from 0 to 100. This approach is widely adopted in the literature, and is associated with attainable yield.[11][8][9][10]

Specific nutrient suitabilities are derived from Awoonor *et al.* (2023) and cross-referenced against Linda and Oluwatola (2015), as is mean maximum temperature, cross-referenced against with Zabe *et al.* (2025).[8] As noted by Waynyama *et al.* (2019), there is an absence of literature on the impact of elevation on maize yield but specific seeds come with elevation recommendations; it is therefore considered sensible to integrate elevation only into the final seed recommendation.

However, there is a clear limitation thus far in that these are global parameters. To improve relevance to Kenya — insofar as data availability permits — KALRO's *Soil Suitability Evaluation for Maize Production in Kenya* (NAAIAP, 2014) is used to finetune the global parameters.[12] These provide "critical levels" for a variety of nutrients; while

Table 1: Land Suitability Classification, based on framework of Sys *et al.* 1993.[11] Attainable yield scores based on Wanyama *et al.* (2019).[9]

Suitability Class	Suitability Score	Description	Attainable Yield
S1(1)	1.0	Highly suitable	95-100%
S1(2)	0.95	Highly suitable	85-95%
S2	0.85	Moderately suitable	60-85%
S2	0.6	Marginally suitable	40-60%
N1	0.4	Not suitable	25-40%
N2	0.25	Not suitable	0-25%

definitions vary, this is generally defined as the nutrient concentration below which crop yield is lower than 90% of optimal yield.[13]

Table 1 details the suitability parameters employed in this study. Suitability for are predominantly derived from from and Awoonor *et al.* (2023), respectively, before being cross-checked against other existing references. [11][8][14][5][15] For climate threshold constraints, maize-specific parameters from Zabel *et al.* (2025) are used, again originally derived from [11].

While suitability data does not exist for zinc and iron, they are considered essential nutrients in maize growth.[16] As such, suitability values for these nutrients, as well as for bulk density, are inferred from existing specialised literature.[17][18][19] However, without empirical validation, these are not employed as strict constraints in the fuzzy logic model detailed in Section 3. While maize growth is known to show only moderate tolerance to boron and is inhibited by Al^{3+} ions at lower pHs, suitability data is not available and so the possible limitations of ion concentrations are not accounted for in the model; bulk density, zinc and iron concentration are excluded for the same reason.[11] Similarly, while gypsum concentration is included as a parameter by Sys *et al.*, its absence in global soil datasets mean it is not considered here.[11]

+ward/ county shape file sources

Bulk density is not included because it is .. (no - can actually do something about it) Zinc and iron are considered important (Major deficiencies in the 4 Counties were N, SOC, P and Zn, followed by K and Mg; Micronutrients, Cu and Fe adequate except in Bomet and Kakamega, respectively. (NAAIAP presentation), so critical level from NAAIAP is given as S1 and, to minimise assumptions, below that is given as S2, but marked as possible Zn/Fe deficiency and testing recommended.

Table 2: Land suitability parameters and thresholds for maize production

Parameter	Data Source	Values	Suitability	Suitability Source
Climate and Growing Conditions				
Growing cycle	–	90–130 days	–	[11]
Mean Season Rainfall (mm)	CHIRPS	0, 300, 400, 500, 600, 750, 900, 1200, 1600	0.25, 0.6, 0.85, 0.95, 1, 1, 0.95, 0.85, 0.60	[11]
Precipitation (1st month, mm)	CHIRPS	0, 60, 75, 100, 125, 175, 220, 295, 400, 475	0.25, 0.6, 0.85, 0.95, 1, 1, 0.95, 0.85, 0.6, 0.25	[11]
Precipitation (2nd month, mm)	CHIRPS	0, 70, 120, 150, 175, 200, 235, 310, 400, 475	0.25, 0.6, 0.85, 0.95, 1, 1, 0.95, 0.85, 0.6, 0.25	[11]
Precipitation (3rd month, mm)	CHIRPS	0, 70, 120, 150, 175, 200, 235, 310, 400, 475	0.25, 0.6, 0.85, 0.95, 1, 1, 0.95, 0.85, 0.6, 0.25	[11]
Precipitation (4th month, mm)	CHIRPS	0, 60, 80, 100, 125, 165, 210, 285, 400, 475	0.25, 0.6, 0.85, 0.95, 1, 1, 0.95, 0.85, 0.6, 0.25	[11]
Mean Temperature (°C)	ERA5	0, 14, 16, 18, 22, 24, 26, 32, 35, 40	0.25, 0.6, 0.85, 0.95, 1, 1, 0.95, 0.85, 0.6, 0.25	[11]
Mean Germination Temperature (°C)	ERA5	0, 10, 13, 18, 21, 24, 29	0, 0.4, 0.85, 1, 0.85, 0.4, 0.25	[11]
Mean Minimum Temp (°C)	ERA5	0, 7, 9, 12, 16, 17, 18, 24, 28, 30	0.25, 0.6, 0.85, 0.95, 1, 1, 0.95, 0.85, 0.6, 0.25	[11]
Mean Maximum Temp (°C)	ERA5	24, 26, 32, 35, 40	1.0, 0.95, 0.85, 0.60, 0.25	[8][5]
Relative Humidity (development stage (2nd month) %)	ERA5	0, 30, 36, 42, 50, 65, 80	0.25, 0.6, 0.85, 0.95, 1, 1, 0.95	[11]

Parameter	Data Source	Values	Suitability	Suitability Source
Relative Humidity (maturation stage (4th month) %)	ERA5	0, 20, 24, 30, 40, 50, 75, 90	0.6, 0.85, 0.95, 1, 1, 0.95, 0.85, 0.6	[11]
Crop Failure Frequency	ERA5	0.0, 0.025, 0.05, 0.075, 0.1, 0.125, 0.15, 0.175, 0.2, 0.225, 0.25	1.0, 0.98, 0.95, 0.88, 0.73, 0.5, 0.27, 0.12, 0.05, 0.02, 0.0	[5][11]
CropSuite Threshold Constraints				
Temp Limits (Low / High, °C)	ERA5	Low: 13.5, High: 39.24	–	[11][5]
Lethal Temp (Min)	ERA5	Value: 5, Duration: 7 days	–	[11][5]
Lethal Temp (Max)	ERA5	Value: 32, Duration: 5 days	–	[11][5]
Lethal Prec (Min)	ERA5	Value: 1, Duration: 21 days	–	[11][5]
Lethal Prec (Max)	ERA5	Value: 100, Duration: 3 days	–	[11][5]
Terrain				
Slope (%)	Copernicus DEM	0, 4, 8, 16, 30, 50	1.0, 0.95, 0.85, 0.60, 0.40, 0.25	[11]
Drainage	iSDA	Good, Moderate, Somewhat poor, Poor (drainable), Poor (not)	1.0, 0.95, 0.85, 0.60, 0.40	[8][11]
Soil Physical Properties				
Soil Texture	iSDA	1,2,3,4,5,6,7,8,9,10,11,12	0.83, 0.25, 0.85, 1.0, 1.0, 0.95, 0.95, 1.0, 0.85, 1.0, 0.77, 0.6	[11][5]
Bulk Density (g cm^{-3})	iSDA	0.80, 1.00, 1.10, 1.30, 1.40, 1.55, 1.70	0.85, 0.95, 1.00, 1.00, 0.95, 0.6, 0.25	Inferred from [17]
Coarse Fragments (vol%)	iSDA	0, 3, 15, 35, 55	1.0, 0.95, 0.85, 0.60, 0.25	[11][8]
Soil Depth (cm)	iSDA	0, 20, 50, 75, 100	0.25, 0.6, 0.85, 0.95, 1.0	[11]
Soil Chemical Properties				
pH (H_2O)	iSDA	0, 5.2, 5.5*, 7.0, 7.8, 8.2, 8.5	0.4, 0.6, 1, 0.95, 0.85, 0.6, 0.25	[11]
Apparent CEC ($cmol_{(+)}$/kg clay)	iSDA	0, 10, 16, 24	0.4, 0.85, 0.95, 1.0	[11][8]
Effective CEC ($cmol_{(+)}$/kg soil)	iSDA	0, 5, 10	0.85, 0.95, 1.0	[20]
Base Saturation (%)	iSDA	0, 20, 35, 50, 80	0.4, 0.6, 0.85, 0.95, 1.0	[11][8]
Organic Carbon (%)	iSDA	0, 0.5, 0.7, 1.5, 2.7*	0.4, 0.6, 0.85, 0.95, 1.0	[11][5]
Total Nitrogen (%)	iSDA	0, 0.1, 0.15, 0.2	0.6, 0.85, 0.95, 1.0	[8]
Available P (mg kg^{-1})	iSDA	0, 6, 30*	0.4, 0.6, 1.0	[8]
Exchangeable Ca ($cmol_{(+)}$/kg)	iSDA	0, 1, 2*	0.6, 0.85, 1.0	[8][14]
Exchangeable K ($cmol_{(+)}$/kg)	iSDA	0, 0.1, 0.24*	0.85, 0.95, 1.0	[8][14]
Exchangeable Mg ($cmol_{(+)}$/kg)	iSDA	0, 0.2, 0.5, 1*	0.25, 0.4, 0.6, 1.0	[8][14]
Zn (mg/kg)	iSDA	0, 5.0*	0.95, 1.0	Inferred from [18][19]
Fe (mg/kg)	iSDA	0, 10*	0.95, 1.0	Inferred from [19][21]
Exchangeable Na (($cmol_{(+)}$/kg)	Africa SoilGrids	0, 0.2, 0.4, 0.5	0.6, 0.85, 0.95, 1	[8]
ECE (dS m^{-1})	Global Salinity	0, 2, 4, 6, 8, 12	1.0, 0.95, 0.85, 0.6, 0.4, 0.25	[8] [11]
ESP topsoil (%)	Africa SoilGrids	0, 8, 15, 20, 25	1.0, 0.95, 0.85, 0.6, 0.25	[11]

2.2.1 Practical Points

Unit Conversions

Unit conversions of iSDA must be carefully treated, as many are stored in log form.

— CHIRPS: No conversion (mm)

— ERA5: Kelvin must be converted to Celsius during extraction from GEE ($T(°C) = T(K) - 273.15$)

— Copernicus DEM: Slope is natively in degrees, so must be converted to % via

$$\text{Slope (\%)} = \tan\left(\text{Slope in Degrees} \times \frac{\pi}{180}\right) \times 100$$

— iSDA data:

- pH: Back transform with x/10.[22][1]
- Clay Content (%): No transform required.[23]
- Apparent CEC (cmol+/kg clay):
- Effective CEC (cmol+/kg soil): Use 20-50cm.[11] Back transform with $exp(x/10)-1$ to obtain cmol+/kg.[24]
- Organic Carbon (%): Use 0-20cm.[11] Back transform with $exp(x/10) - 1$[25] to obtain g/kg[22]; divide by 10 to obtain %.[1]
- Total Nitrogen (%): Back transform with $exp(x/100) - 1$[26] to obtain g/kg[1]; divide by 10 to obtain %.
- Extractable Phosphorus (mg/kg): Back transform with $exp(x/10) - 1$ to obtain mg/kg.[27]
- Extractable Calcium (cmol+/kg): Back transform with $exp(x/10) - 1$[28] to obtain mg/kg[1];

$$\text{Ca } (\text{cmol}_c/\text{kg}) = \frac{\text{Ca (mg/kg)} \times 2}{40.078 \times 10}$$

- Extractable Potassium (cmol+/kg): Back transform with $exp(x/10) - 1$[29] to obtain mg/kg[1];

$$\text{K } (\text{cmol}_c/\text{kg}) = \frac{\text{K (mg/kg)} \times 1}{39.098 \times 10}$$

- Extractable Magnesium (cmol+/kg): Back transform with $exp(x/10) - 1$[30] to obtain mg/kg[1];

$$\text{Mg } (\text{cmol}_c/\text{kg}) = \frac{\text{Mg (mg/kg)} \times 2}{24.305 \times 10}$$

- Extractable Zinc (mg/kg): Back transform with $exp(x/10) - 1$[31] to obtain mg/kg[1]
- Extractable Iron (mg/kg): Back transform with $exp(x/10) - 1$[32] to obtain mg/kg[1]
- Stone Content (%): Back transform with $exp(x/10) - 1$[33] to obtain percentage.
- Depth to Bedrock: No transform required.[34]
- Soil Texture: No transform required.[35]

  The apparent CEC is defined by Sys *et al.* as Apparent Cation Exchange Capacity, defined by Sys *et al.* (1993) as,

$$CEC_{clay} = \frac{\text{Total Mineral CEC}}{\text{\% Clay}} \times 100 = \frac{\text{Effective CEC} - (2.6 \times \%OrganicCarbon)}{\%Clay} \times 100$$

  is also derived from the iSDA data.[11]

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Texture Classes

Description	Value (iSDA)	Value (Sys *et al.* (1993))	Suitability
Clay	1	Cm; C+60,v; C+60,s; C-60,v; C-60,s; Co	0.83
Silty Clay	2	SiCm, SiCs	0.25
Sandy Clay	3	SC	0.85
Clay Loam	4	CL	1.0
Silty Clay Loam	5	SiCL	1.0
Sandy Clay Loam	6	SCL	0.95
Loam	7	L	0.95
Silt Loam	8	SiL	1.0
Sandy Loam	9	SL	0.85
Silt	10	Si	1.0
Loamy Sand	11	LfS; LS; LcS	0.77
Sand	12	fS; S	0.6

Table 3: Mapping between soil texture descriptions, iSDA classification values, classification symbols from Sys *et al.* (1993), and suitability (averaged e.g. where Clay is split over different classes - no doubt an issue considering the unsuitability of massive clay[11]).

2.2.2 Considerations of sodicity and salinity

Electrical conductivity (EC_e) and exchangeable sodium are also considered as possible parameters, which may be obtained from the Global Soil Salinity Map (Ivushkin *et al.*, 2019) and Africa SoilGrids (Hengl *et al.*, 2015) respectively, both at 250m resolution.[36][37]Exchangeable Sodium Percentage (ESP) and Base Saturation (BS) may then be derived as below,

$$ESP = \frac{C_{Na^+}}{CEC} \times 100 \qquad BS = \frac{\sum(C_{Ca^{2+}}, C_{Mg^{2+}}, C_{K^+}, C_{Na^+})}{CEC} \times 100$$

where C_{Na^+} is the concentration of exchangeable sodium ($cmol_c/kg$), CEC is the Cation Exchange Capacity ($cmol_c/kg$), and $\sum(C_{Ca^{2+}}, C_{Mg^{2+}}, C_{K^+}, C_{Na^+})$ is the sum of the basic exchangeable cations ($cmol_c/kg$).[38][39]

However, Africa SoilGrids does not offer uncertainty maps, only providing a summary RMSE for each soil parameter (3.61 for exchangeable sodium).[37] Considering the range in Nandi is ∼6 cmol+/kg, this is unreliable. We include in Table 2 details on the distribution of exchangeable sodium concentrations in Nandi. Given the range of approximately 6 cmol+/kg, a RMSE of 3.61 risks introducing a high degree of percentage error into the downstream suitability aggregation.

Likewise, the Global Soil Salinity map, which classifies EC_e (ds/m) into five classes (Non-saline (<2), Slightly (2-4), Moderately (4-8), Highly (8-16), and Extremely (>16)), does not provide an uncertainty measure.[36] Considering this and the fact that all measured soil in Nandi falls into the non-saline class (with associated suitability of 1.0) - and thus sodium concentration will never be a limiting factor - salinity is not included in the overall suitability calculation. This also means base saturation is not included, but it is considered that this is already represented in the suitabilities of the individual basic ions.

Statistic	Value (cmol$^+$/kg)
Minimum Value	0.0001
Maximum Value	6.0200
Mean Value	0.4647
Standard Deviation	0.3668

Table 4: Summary Statistics for Nandi Sodium (ENAX)

Suitability values are generally defined for CEC_clay.[11] However, this leads to inflation at low clay percentages and undefined values at zero. For the 612 sites in Nandi where Clay % is below 10% (here, for 612 sites with Clay % = 0), suitability values for CEC_soil are used, with suitability values derived from the critical values presented in Adamu *et al.* (2015).

2.2.3 Visualisation

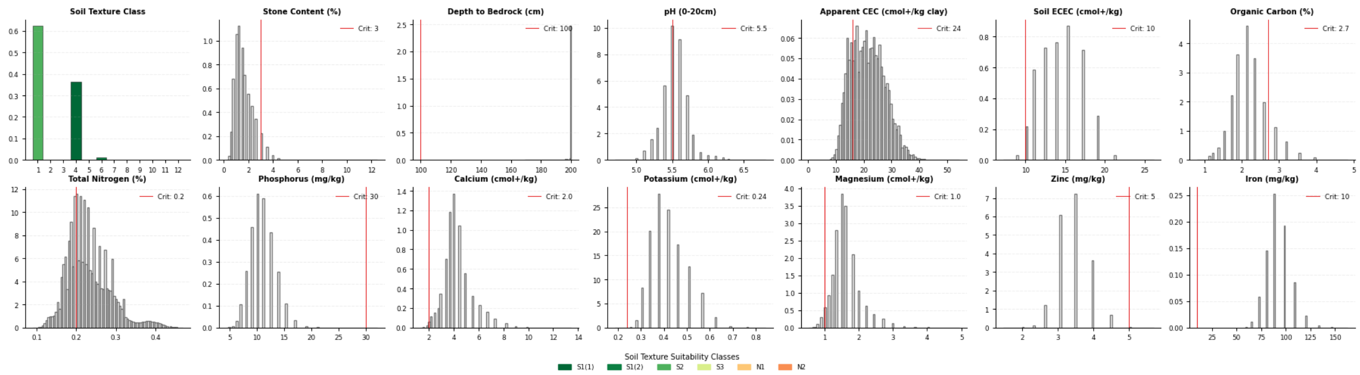

Figure 1: Distribution of soil physical and chemical properties in Nandi County. Soil Texture Class colors represent suitability based on a gradient from red (low suitability) to green (high suitability).

Figure 1 allows validation of the data against empirical knowledge. Nandi soils are rich in iron (in agreement with (NAAIAP, 2014) as well as magnesium, potassium, and calcium, also leading to the majority of soils being above the critical threshold for Apparent and Effective CEC. However, there are clear deficiencies in phosphorus and zinc, as well as (but less stark) organic carbon and total nitrogen (again in agreement with NAAIAP, 2014). Physical-properties wise, bedrock is sufficiently deep for maize production and texture classes are almost entirely favourable; however, the majority of soil is affected by stone content.

2.2.4 Depth Weighting

Sys *et al.* (1991) note differing effects of soil properties on growth depending on depth, suggesting weighting factors for different soil depths.[11][5] iSDA soil data provides two different depths, 0-20cm and 20-50cm[1], for which *Sys et al.* (1991) propose weighting factors of 2 and 1.5 respectively in the general case.[11] This is caveated for specific features:

— When coarse fragments (stone content) decrease with depth, the mean content is used; however, in cases where content increases or is constant, standard weighting is suggested,[11]

— Apparent CEC (and Effective CEC) should use the B horizon (50 cm) if soils are uniform, and a weighted average up to 75cm if profiles have rejuvenated topsoil.[11]

— For pH, cations, and organic carbon, a depth of 0-25cm should be used.[11]

— Profiles with heterogeneous textures require weighting.[11]

As such, we analyse stone content and texture heterogeneity over the two depths.

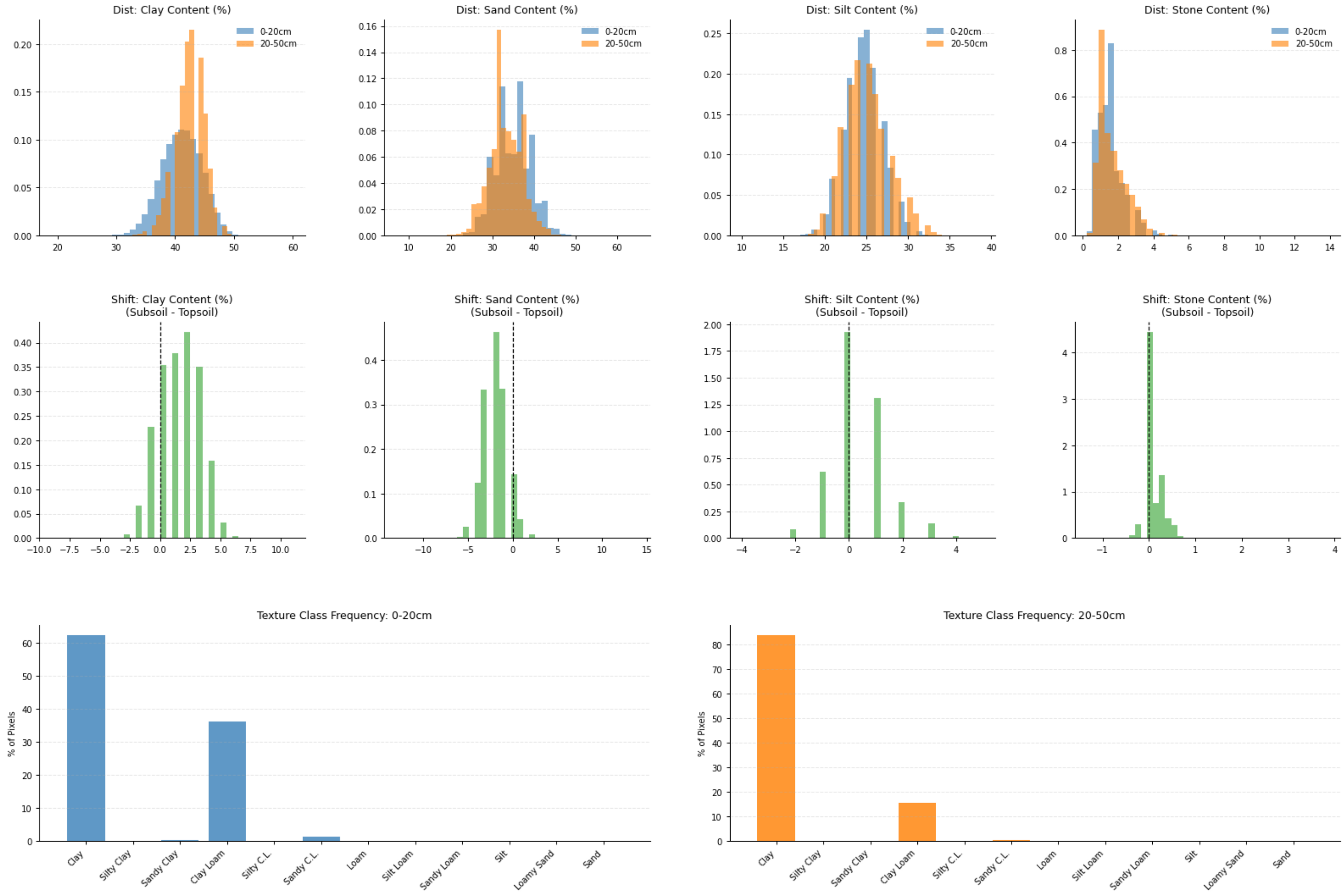

Figure 2: Stone content and texture heterogeneity over 0-20cm and 20-50cm depths.

From Figure 2, stone content is roughly homogeneous — marginally increasing - between the two depths; as such, standard weighting is employed. The same is done for texture, which exhibits a clear increase in clay content compared to sand with depth, with silt content staying approximate constant.

stone content

3 Methods

3.1 Conceptual Framework

Crop suitability assessment is conducted using Liebig's Law of the Minimum [? 11], extended to incorporate spatially explicit measurement uncertainty from continental-scale soil predictions via Monte Carlo simulation. The framework addresses a critical gap in land evaluation under remote sensing constraints: deterministic models that use predicted rather than measured soil properties ignore prediction uncertainty, potentially producing overconfident suitability classifications in data-sparse regions.

Let x_i^p denote the true (but unknown) value of environmental variable i at pixel p, and let $\mu_i(x_i)$ represent the corresponding suitability membership function mapping physical values to scores in $[0, 1]$, where 0 indicates complete unsuitability and 1 represents optimal conditions. Under Liebig's principle, overall suitability is determined by the single most limiting factor:

$$S^p = \min_{i \in \mathcal{I}} \mu_i(x_i^p) \tag{1}$$

where $\mathcal{I}$ is the set of environmental variables considered. This formulation reflects the biological reality that crop growth is constrained by whichever resource or condition is most deficient, with limited scope for compensation between factors [? ?]. This is distinct from multiplicative or geometric mean aggregation schemes, which allow partial compensation between favourable and unfavourable conditions; such compensation is biologically unrealistic for limiting-factor constraints such as soil pH or waterlogging.

When x_i^p is a prediction rather than a direct measurement, uncertainty in the true value propagates through the suitability assessment. We model this via Monte Carlo sampling, computing the expected suitability as:

$$\bar{S}^p = E\left[\min_{i \in \mathcal{I}} \mu_i(x_i^p)\right] = \frac{1}{N}\sum_{n=1}^{N}\left[\min_{i \in \mathcal{I}} \mu_i(x_i^{p,(n)})\right] \tag{2}$$

where $x_i^{p,(n)}$ is the n-th sample from the uncertainty distribution for variable i at pixel p, and N is the number of Monte Carlo iterations. Critically, $E[\min_i \mu_i] \neq \min_i E[\mu_i]$ due to Jensen's inequality: the expected minimum is systematically lower (more conservative) than the minimum of expected values. Deterministic suitability mapping using predicted soil means therefore produces an optimistic bias whose magnitude depends on both the prediction uncertainty and the slope of the membership function at that value. Monte Carlo estimation of $E[\min_i \mu_i]$ corrects this bias directly.

Primary structural limitation: variable independence. All variables are sampled independently across Monte Carlo iterations. This is a significant structural assumption. Known correlations exist between soil properties: pH covaries with base saturation, calcium, and magnesium; clay content covaries with CEC and moisture retention; temperature and precipitation are temporally correlated within seasons. Independent sampling will periodically generate physically implausible combinations (e.g. low pH with very high base saturation), which can distort the minimum operator in either direction. iSDAsoil does not provide cross-variable covariance matrices at pixel level, making formal copula sampling infeasible as a primary approach. The consequences of this assumption are assessed via a Gaussian copula sensitivity analysis (Section 3.5.5), in which a Spearman correlation matrix estimated empirically from iSDAsoil predicted values across Nandi County is used to generate correlated joint samples. This bounds the degree to which the independence assumption affects suitability classifications in the study region.

3.2 Variable-Level Uncertainty Distributions

3.2.1 Soil Variables

Suitability for soil variables is assessed using the 0–20 cm iSDAsoil layer only. Smallholder maize rooting depth in tropical highland soils is predominantly confined to the upper 20–30 cm, particularly where compaction layers, plough pans, or high-clay subsoil horizons restrict root penetration [? ?]. The 0–20 cm layer captures the primary root zone where nutrient uptake, pH sensitivity, and physical constraints are agronomically most consequential for annual smallholder crops. This departs from the generic two-layer depth-weighting scheme of Sys et al. [11], which was developed for a broader range of crop types and soil conditions. The departure is further supported by the very strong vertical autocorrelation observed between the 0–20 cm and 20–50 cm iSDA layers across Nandi County ($\rho > 0.90$ for all variables), meaning the deeper layer adds minimal independent information beyond what is already captured at the surface.

For each soil variable i with iSDAsoil predicted value $\hat{x}_i^p$ and published 5th–95th percentile prediction interval $[\hat{x}_i^{(5),p}, \hat{x}_i^{(95),p}]$ [1], the appropriate uncertainty distribution depends on the distributional properties of the variable.

Distribution selection. Soil properties differ substantially in distributional form. Percentage-based variables with approximately symmetric distributions (sand, silt, clay content; pH; bulk density) are modelled as truncated Gaussian distributions, reflecting symmetric uncertainty around the predicted mean. However, several agronomically important variables are known to be strongly right-skewed at continental scales: organic carbon, extractable phosphorus, zinc, iron, and potassium. For these variables, a truncated log-normal distribution is used:

$$\ln(x_i^p) \sim \mathcal{TN}(\mu = \ln(\hat{x}_i^p),\ \sigma = \sigma_i^p,\ a = \ln(L_i),\ b = \ln(U_i)) \tag{3}$$

where σ_i^p is estimated from the published prediction interval:

$$\sigma_i^p = \frac{\ln(\hat{x}_i^{(95),p}) - \ln(\hat{x}_i^{(5),p})}{2 \times 1.645} \tag{4}$$

For symmetric variables, a truncated Gaussian is used instead:

$$x_i^p \sim \mathcal{TN}(\mu = \hat{x}_i^p,\ \sigma = \sigma_i^p,\ a = L_i,\ b = U_i) \tag{5}$$

$$\sigma_i^p = \frac{\hat{x}_i^{(95),p} - \hat{x}_i^{(5),p}}{2 \times 1.645} \tag{6}$$

In both cases, $[L_i, U_i]$ represents the physically plausible range for variable i (Table 2). Truncation ensures samples respect natural bounds. The choice of distribution family is consequential under the minimum operator: Liebig suitability is sensitive to the lower tail of each variable's distribution, and a Gaussian approximation to a right-skewed variable will systematically underestimate lower-tail probability, producing an optimistic bias in expected suitability. The log-normal parameterisation avoids this. The sensitivity of classifications to this choice is evaluated in Section 3.5.4.

Reliability stratification via CCC. Variables were stratified by prediction reliability using published Concordance Correlation Coefficients [CCC; ?] from iSDAsoil's continental validation (Table 3). The CCC penalises both imprecision and systematic bias, making it more appropriate than Pearson correlation for prediction–observation agreement [1]. Three tiers are defined:

- **High-reliability (CCC $\geq$ 0.85):** pH (0.900), organic carbon (0.883), bulk density (0.901), ECEC (0.860), extractable Ca (0.913), Mg (0.898), K (0.872), Fe (0.899), Al (0.937). These variables enter the minimum aggregation with their native uncertainty distributions.
- **Medium-reliability (0.75 $\leq$ CCC $<$ 0.85):** Sand (0.848), clay (0.854), silt (0.780), total nitrogen (0.845), zinc (0.831), total carbon (0.820). These variables are included in the minimum aggregation, with outputs explicitly flagged for moderate prediction uncertainty.
- **Low-reliability (CCC $<$ 0.75):** Extractable phosphorus (0.654), sulphur (0.708), stone content (0.701), depth to bedrock (0.725). These variables are included in the Liebig minimum aggregation using their published iSDAsoil prediction intervals without modification. The published intervals may underestimate true uncertainty for these variables, since systematic prediction bias — which the CCC penalises — is not necessarily reflected in interval width. Results for low-reliability variables are therefore reported as advisory outputs, presented in supplementary maps distinct from the main suitability classification, and interpreted with explicit caution. This is preferable to exclusion, which would bias suitability upwards by ignoring potentially limiting factors: extractable phosphorus in particular is a well-documented constraint in sub-Saharan African soils [?].

The CCC thresholds (0.75 and 0.85) were selected based on natural clustering in the published validation statistics and align with conventional CCC interpretation (excellent: >0.90; good: 0.75–0.90; moderate: 0.50–0.75; ?). Threshold sensitivity is evaluated in Section 3.5.3.

3.2.2 Terrain Variables

Slope, derived from the Copernicus GLO-30 DEM at 30 m resolution, is treated as deterministic. DEM-derived slope uncertainty is typically sub-degree at this resolution [?], which translates to negligible suitability variation relative to the membership function breakpoints in Table 1 (e.g., the breakpoint from S1 to S2 suitability occurs at 8%, corresponding to a slope change of $\sim$4.6°). Treating slope as deterministic is therefore a reasonable approximation that does not materially affect suitability uncertainty. Drainage class is modelled as a discrete ordinal variable with fixed suitability scores by class (Table 1); no distributional uncertainty is assigned.

3.2.3 Climate Variables

Climate suitability incorporates both mean seasonal conditions and extreme event frequency. Mean seasonal temperature and precipitation over the 110-day growing season (2015–2024 average) were computed from ERA5-Land ($\sim$9 km resolution) and CHIRPS ($\sim$5 km resolution) daily data, respectively (Section 2.1). Suitability scores for mean conditions, μ_{temp} and μ_{precip}, are derived from piecewise-linear membership functions calibrated to maize physiological thresholds [Table 1; 11]. These mean climate terms are treated as fixed across Monte Carlo iterations; interannual variability in mean seasonal conditions is implicitly captured by the ten-year averaging period.

Crop failure frequency with ENSO conditioning. Crop failure frequency for each pixel was determined from year-by-year analysis of CropSuite lethal threshold violations [5]:

- Lethal minimum temperature: <5°C for 7 consecutive days
- Lethal maximum temperature: >32°C for 5 consecutive days
- Lethal drought: <1 mm precipitation for 21 consecutive days
- Lethal flooding: >100 mm precipitation in 3 consecutive days

A pixel was classified as a crop failure in year y if any threshold was violated during the growing season window.

Rather than restricting failure frequency estimation to a ten-year window — which is too short to separate ENSO-phase effects from sampling noise — the full available ERA5-Land and CHIRPS record is used (1981–2024 for the Long Rains season; 1981–2024 for the Short Rains season, covering $T = 43$ and $T = 43$ growing seasons respectively). Each year is classified by ENSO phase using the NOAA Oceanic Niño Index (ONI): El Niño (ONI $\geq +0.5$), La Niña (ONI ≤ -0.5), or Neutral, averaged over the primary teleconnection months for each season (March–May for Long Rains; October–December for Short Rains) [?]. This yields observed failure counts k_s^p within n_s years of phase $s \in \{\text{EN}, \text{LN}, \text{N}\}$ for each pixel.

Under independent uniform Beta(1,1) priors for each phase, the Bayesian posteriors are:

$$p_s^p \sim \text{Beta}(1 + k_s^p,\ 1 + n_s - k_s^p)\,, \quad s \in \{\text{EN, LN, N}\} \tag{7}$$

Long-run ENSO phase frequencies $\boldsymbol{\pi} = (\pi_{\text{EN}}, \pi_{\text{LN}}, \pi_{\text{N}})$ are estimated from the full ONI record (1950–2024), giving stable empirical phase probabilities across many ENSO cycles.

For each Monte Carlo iteration n, climate failure suitability is computed by first drawing an ENSO state $s^{(n)} \sim$ Categorical($\boldsymbol{\pi}$), then drawing a phase-conditional failure probability:

$$p^{p,(n)} \sim \text{Beta}\big(1 + k^p_{s^{(n)}},\ 1 + n_{s^{(n)}} - k^p_{s^{(n)}}\big) \tag{8}$$

This is converted to a reliability suitability score via the crop failure frequency membership function (Table 1):

$$\mu_{\text{rel}}^{p,(n)} = \mu_{\text{rel}}\left(p^{p,(n)}\right) \tag{9}$$

This formulation correctly marginalises over both ENSO phase uncertainty and within-phase binomial sampling uncertainty. It addresses the key structural problem of the ten-year approach: failure years in East Africa are positively autocorrelated through ENSO, so pooling across phases without conditioning conflates climate risk with climatological sampling bias. A pixel that experienced two La Niña years out of a ten-year window dominated by neutral or El Niño years would have its failure probability substantially underestimated under the unconditional model; ENSO conditioning resolves this. With $n_s \approx 13$–17 observations per phase over the extended record, the phase-conditional posteriors are reasonably well-constrained while remaining appropriately uncertain.

The final climate suitability for iteration n combines mean seasonal adequacy and extreme event reliability under Liebig's minimum:

$$\mu_{\text{climate}}^{p,(n)} = \min\left(\mu_{\text{temp}}^{p},\ \mu_{\text{precip}}^{p},\ \mu_{\text{rel}}^{p,(n)}\right) \tag{10}$$

This formulation treats chronic unsuitability (inadequate mean conditions) and acute failure risk (extreme events) as equally subject to Liebig's principle of limitation.

Scale mismatch note. Climate data (ERA5: ~9 km; CHIRPS: ~5 km) are downscaled to 30 m to match the soil and DEM resolution. This downscaling does not physically resolve sub-pixel climate variation; spatial patterns within each coarse pixel are inherited from the resampling algorithm rather than from observed meteorological gradients. Climate-driven suitability variation at scales below ~5–9 km should therefore be interpreted with caution. The fraction of total pixel-level suitability variance attributable to within-climate-cell variation versus between-cell variation is quantified in the Results (Section 4.x), providing an empirical assessment of how much discriminatory information the fine-resolution soil and terrain data contribute relative to the coarser climate inputs.

3.3 Monte Carlo Aggregation and Convergence

For each pixel p, expected suitability under measurement uncertainty is computed via Monte Carlo sampling with $N = 100$ iterations. The full per-iteration suitability is computed as the Liebig minimum across all variables:

$$S^{p,(n)} = \min\left(\mu_{\text{climate}}^{p,(n)},\ \min_{i\in\text{soil}} \mu_i^{p,(n)},\ \min_{i\in\text{terrain}} \mu_i^{p,(n)}\right) \tag{11}$$

Expected suitability and uncertainty are then:

$$\bar{S}^p = \frac{1}{N}\sum_{n=1}^{N} S^{p,(n)} \tag{12}$$

$$\sigma_S^p = \sqrt{\frac{1}{N-1}\sum_{n=1}^{N}\left(S^{p,(n)} - \bar{S}^p\right)^2} \tag{13}$$

$N = 100$ was selected following convergence analysis targeted at pixels near suitability class boundaries, where convergence is slowest and misclassification risk is highest. Pixels where $\bar{S}^p \in [0.23, 0.27]$ or $\bar{S}^p \in [0.48, 0.52]$ — straddling the S3/N and S2/S3 thresholds respectively — were identified and mean suitability estimates and classification stability (fraction of iterations assigned to each class) were monitored as a function of N from 10 to 500. Estimates stabilised within 0.01 units and classification stability exceeded 95% by $N = 100$ for all boundary pixels examined; increasing to $N = 500$ produced changes <0.005. Convergence diagnostics are reported in Supplementary Figure S1. Computational cost scales linearly with N; $N = 100$ represents a balance between precision and feasibility at regional scale (Nandi County: ~3.2 million pixels at 30 m resolution, requiring approximately 12 minutes on an NVIDIA RTX 3080 GPU with fully vectorised NumPy operations).

3.4 Output Classification and Confidence Tiers

Pixels were classified into four suitability classes following FAO conventions [? 11]:

- **Highly Suitable (S1):** $\bar{S}^p > 0.75$. Minimal limitations for maize production under current management.
- **Moderately Suitable (S2):** $0.50 < \bar{S}^p \leq 0.75$. Moderate limitations requiring management intervention.
- **Marginally Suitable (S3):** $0.25 < \bar{S}^p \leq 0.50$. Severe limitations; marginal viability under smallholder conditions.
- **Unsuitable (N):** $\bar{S}^p \leq 0.25$. Prohibitive limitations for maize production.

Each pixel is additionally assigned a confidence tier based on the Monte Carlo standard deviation σ_S^p:

- **High confidence:** $\sigma_S^p < 0.10$. Narrow uncertainty; suitability classification is reliable.
- **Medium confidence:** $0.10 \leq \sigma_S^p < 0.20$. Moderate uncertainty; classification probable but not definitive.
- **Low confidence:** $\sigma_S^p \geq 0.20$. Wide uncertainty; classification spans multiple classes within the credible interval.

This two-dimensional output distinguishes a well-constrained moderate suitability estimate (e.g., $\bar{S} = 0.65 \pm 0.08$, High Confidence) from a highly uncertain one assigned the same nominal class (e.g., $\bar{S} = 0.67 \pm 0.22$, Low Confidence). Low-confidence pixels identify locations where additional ground-truth soil data collection would most reduce classification uncertainty. Advisory outputs for low-reliability variables (Section 3.2.1) are presented as supplementary maps rather than incorporated into the primary classification, allowing practitioners to assess P and stone content constraints without contaminating the main uncertainty-calibrated results.

3.5 Validation and Sensitivity Analysis

3.5.1 Validation Against FAO-GAEZ

Predicted suitability scores are compared to FAO Global Agro-Ecological Zones (GAEZ v4) maize suitability estimates for Nandi County [?]. GAEZ provides independent suitability assessments based on the FAO Land Evaluation framework with distinct data sources and deterministic aggregation methods. Spearman rank correlation ρ_s is computed to assess ordinal agreement in spatial patterns; absolute differences in mean suitability score quantify the direction and magnitude of any systematic divergence. Regions of disagreement are analysed to identify where uncertainty-aware probabilistic methods produce divergent classifications from GAEZ's deterministic approach, and to assess whether divergences are associated with high-σ_S pixels (suggesting uncertainty rather than systematic bias as the cause of differences).

3.5.2 Aggregation Method Comparison

To characterise the sensitivity of results to the choice of aggregation operator, a geometric mean aggregator is implemented as a contrast to the Liebig minimum:

$$S_{\text{geom}}^{p,(n)} = \left(\prod_{i=1}^{m} \mu_i^{p,(n)} \right)^{1/m} \tag{14}$$

Both operators are applied to a 10 km × 10 km test region within Nandi County. Classification agreement is quantified via confusion matrix; pixels where the two operators produce different suitability classes are mapped and examined for common characteristics (e.g. presence of a single strongly limiting factor near a threshold). The geometric mean permits partial compensation between factors and is appropriate if the biological reality is that favourable conditions in one dimension can partially offset deficiencies in another; the Liebig minimum assumes no such compensation. This comparison directly tests how consequential this modelling choice is for the study region.

3.5.3 CCC Threshold Sensitivity

The impact of CCC stratification thresholds on classification results is tested by varying the high/medium boundary over $\{0.80, 0.85, 0.90\}$ and the medium/low boundary over $\{0.70, 0.75, 0.80\}$. For each of the nine resulting threshold combinations, classification agreement (percentage of pixels with identical suitability class) is computed against the baseline configuration. This tests whether the tier assignments — and consequently which variables are treated as advisory rather than primary — materially affect suitability classifications across Nandi County. Results are reported in Supplementary Table S1.

3.5.4 Distribution Family Sensitivity

To assess the impact of the distribution family choice for right-skewed variables, suitability maps are generated under two alternative parameterisations for organic carbon, extractable phosphorus, and zinc: (i) truncated Gaussian as used for symmetric variables, and (ii) the baseline truncated log-normal. Classification agreement between the two is quantified. This directly tests whether the Gaussian approximation materially biases expected suitability for these variables in the study region, and provides empirical support for the log-normal choice.

3.5.5 Gaussian Copula Sensitivity Analysis

To assess the consequences of the variable independence assumption (Section 3.1), suitability classifications under independent sampling are compared to those under copula-based joint sampling. A Spearman rank correlation matrix $\mathbf{R}$ is estimated from iSDAsoil predicted means across all pixels in Nandi County, capturing the empirical co-variation structure of soil properties at the study site. This matrix is converted to a Gaussian copula correlation matrix via the standard approximation $R_{ij}^{\text{Gauss}} = \sin(\frac{\pi}{2} R_{ij}^{\text{Spearman}})$ [?]. Correlated joint samples are generated in each Monte

Carlo iteration via Cholesky decomposition: a standard normal vector with the copula correlation structure is drawn, then each marginal is mapped through its inverse CDF (truncated normal or log-normal as specified in Section 3.2.1). Marginal distributions are unchanged from the baseline; only the dependence structure differs.

Suitability classifications under copula sampling are compared to the independent baseline via confusion matrix across all Nandi pixels. Pixels where classifications diverge are mapped and examined for co-located multi-variable stress — specifically whether divergence concentrates in regions where pH, base saturation, and exchangeable cations are simultaneously near threshold values, which is the physically expected location of independence-assumption bias. If fewer than approximately 5% of pixels change suitability class, the independence assumption is a named limitation with bounded practical consequences. If divergence is substantially larger, the copula result is reported as the preferred classification and the independent result as the sensitivity case.

3.6 Computational Implementation

All analyses are conducted in Python 3.10 using NumPy (array operations), SciPy (truncated normal and log-normal distributions, Beta distribution), Rasterio (geospatial I/O), and Google Earth Engine (climate data extraction and long-record failure frequency computation). Monte Carlo sampling is vectorised across all pixels simultaneously using NumPy array broadcasting, avoiding explicit pixel-level loops. Truncation is applied using `scipy.stats.truncnorm` and a log-transform wrapper for log-normal variables. Cholesky decomposition for copula sampling uses `numpy.linalg.cholesky`. The full workflow, from raw data extraction to final suitability maps, is documented in the project repository.

Variable	**Unit**	**CCC**	**Tier**
pH in H_2O	–	0.900	High
Organic carbon	g/kg	0.883	High
Bulk density, <2 mm	g/cc	0.901	High
Effective CEC	cmol(+)/kg	0.860	High
Calcium, extractable	mg/kg	0.913	High
Magnesium, extractable	mg/kg	0.898	High
Potassium, extractable	mg/kg	0.872	High
Iron, extractable	mg/kg	0.899	High
Aluminium, extractable	mg/kg	0.937	High
Sand content	%	0.848	Medium
Clay content	%	0.854	Medium
Silt content	%	0.780	Medium
Total nitrogen	g/kg	0.845	Medium
Zinc, extractable	mg/kg	0.831	Medium
Carbon, total	g/kg	0.820	Medium
Extractable phosphorus	mg/kg	0.654	Low (advisory)
Stone content	%	0.701	Low (advisory)
Depth to bedrock	cm	0.725	Low (advisory)
Sulphur, extractable	mg/kg	0.708	Low (advisory)

Table 5: Concordance Correlation Coefficients (CCC) for iSDAsoil variables [1], with reliability tier assignment. High (blue): CCC ≥ 0.85; Medium (grey): $0.75 \leq$ CCC < 0.85; Low/advisory (orange): CCC < 0.75. Low-reliability variables are included in the Liebig minimum aggregation using their published prediction intervals; results are reported as advisory outputs given that interval coverage for these variables may be underestimated.

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